# Design and Evaluation of an Explainable Legal AI for Bangladeshi Traffic Law

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# Abstract

Large Language Models (LLMs) can make legal information more easy to access, but they can also produce convincing answers which lack reliable evidence. This creates serious risks in high-stakes legal contexts, because users may rely on inaccurate or misleading information. This FYDP proposes an explainable Retrieval-Augmented Generation (RAG)-based legal assistant for Bangladeshi statutory law, with an initial focus on the Bangladesh Road Transport Act, 2018.

The project will develop an expert-reviewed legal question–answer dataset in native Bangla and build a RAG pipeline that retrieves relevant statutory provisions before generating answers grounded in those sources. It will also examine whether greater transparency can help users identify and respond appropriately to any hallucinations that remain. To study this, the project will present the same RAG backend through multiple interface designs that provide different levels of source traceability and uncertainty information.

A controlled user study involving both general users and participants with legal-domain knowledge will evaluate how these interface designs affect hallucination detection, reliance on AI-generated answers, cognitive workload, and usability. The project aims to contribute a grounded legal-AI prototype, a reusable evaluation dataset, and empirical evidence that can support the design of safer and more transparent public-facing Legal AI systems in Bangladesh.

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# Publication List

[Optional] The main contributions of this research are either published or accepted or in preparation in journals and conferences as mentioned in the following list:

## Journal Articles

1.

## Conference Papers

1.

## Additional Publications

Following is the list of relevant publications published in the course of the research that is not included in the thesis:

1.

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# Chapter 1

# Introduction

This chapter is an overview of the entire project, including the problem background, motivation, and the main objectives. The proposed methodology, expected outcomes and overall structure of the report are also briefly described.

## 1.1 Project Overview

Large Language Models (LLMs) have significantly improved the accessibility of information by enabling users to obtain answers through natural language interaction. However, despite their capabilities, these systems often generate inaccurate or fabricated information, commonly referred to as hallucinations. The consequences of such errors become particularly critical in high-stakes domains such as law, where incorrect information may lead to legal misunderstandings and poor decision-making.

This project focuses on developing and evaluating a Legal AI system for Bangladeshi statutory law. The system will use a Retrieval-Augmented Generation (RAG) architecture to retrieve relevant information from official legal documents and generate responses grounded in those sources. It will also support user queries in Bangla to make legal information more accessible to Bangla-speaking users.

Beyond building the technical system, the project will investigate how different interface designs affect a user's ability to recognize and respond to AI-generated hallucinations. To study this, the system will include multiple interface variations with different levels of transparency, including features such as source traceability, legal citations, and uncertainty indicators.

The project will evaluate these interface designs through controlled user studies involving both general users and participants with legal expertise. The study will examine factors such as hallucination detection, appropriate reliance on AI-generated responses, cognitive workload, and overall usability.

Through this work, the project aims to contribute to the development of more transparent, trustworthy, and user-centered Legal AI systems for Bangla-speaking users. It will also provide practical and empirical insights for research in Human-Computer Interaction

(HCI), Explainable Artificial Intelligence (XAI), and Legal Technology.

## 1.2 Motivation

This project is motivated by the growing use of AI assistants for accessing legal information, especially in low-resource languages like Bangla. Large Language Models (LLMs) can help users quickly find legal information, but they can also generate incorrect or unsupported answers. In the legal domain, even small mistakes can cause serious misunderstandings because legal information directly affects people's rights and responsibilities.

This project focuses on the Bangladesh Road Transport Act, 2018, as traffic laws impact millions of citizens through licensing, penalties, and compliance requirements. Since many users do not have legal expertise, they may struggle to identify incorrect AI-generated information. Therefore, this project aims to develop a transparent and reliable Legal AI system.

From an engineering perspective, this project focuses on two main areas:

1. **Backend RAG Architecture Motivation**

   This project uses a Retrieval-Augmented Generation (RAG) architecture instead of relying only on model knowledge or expensive fine-tuning. RAG retrieves relevant legal documents before generating responses, helping the system provide answers grounded in authoritative sources. However, RAG does not completely remove hallucinations, especially when dealing with complex legal language and Bangla statutory documents.

   Research on RAG shows that combining language generation with external retrieval can improve factual accuracy while keeping knowledge sources accessible and updateable [1]. However, studies such as RAGTruth, SELF-RAG, and long-context retrieval analysis show that unsupported responses can still occur even with retrieved evidence [2, 3, 4].

2. **Frontend Explainability and Trust Calibration Motivation**

   Since AI systems cannot guarantee perfect accuracy, this project also studies how interface design can help users better understand and evaluate AI responses. Features such as source citations, confidence indicators, uncertainty visualization, and highlighted evidence may improve users' ability to detect incorrect information and build appropriate trust. This project investigates these factors through a controlled Human–Computer Interaction (HCI) study.

This project also addresses the Complex Engineering Problem framework outlined by BAETE by considering technical performance, transparency, safety, and accountability in a critical domain. The proposed system aims to support both citizens and legal professionals by making legal information easier to access and helping users verify AI-generated responses.

## 1.3 Objectives

The main objectives of this project are:

1. **Develop a High-Quality Legal Dataset**

   Create an expert-reviewed benchmark dataset from the *Road Transport Act, 2018* by extracting and organizing relevant legal texts for Bangla legal question-answering.

2. **Develop a RAG-Based Legal Assistant**

   Design a Retrieval-Augmented Generation (RAG)-based system that retrieves relevant legal provisions before generating responses, helping reduce hallucinations and improve factual accuracy.

3. **Evaluate Different AI Models**

   Compare multiple Large Language Models (LLMs), including proprietary models such as GPT and Gemini, as well as local and open-source models, to understand their effectiveness in answering Bangla legal questions.

4. **Study User Trust and System Usability**

   Conduct user studies with general users and legal experts to evaluate trust in AI-generated legal responses, error detection ability, workload, and usability using NASA-TLX and SUS measurements.

## 1.4 Methodology

This study follows a mixed-methods approach that combines the development of a Retrieval-Augmented Generation (RAG)-based legal question-answering system with a controlled Human–Computer Interaction (HCI) experiment. The research consists of two main phases: developing a Legal AI assistant for the *Bangladesh Road Transport Act, 2018*, and studying how interface design affects users' ability to identify AI hallucinations and build appropriate trust in AI-generated responses.

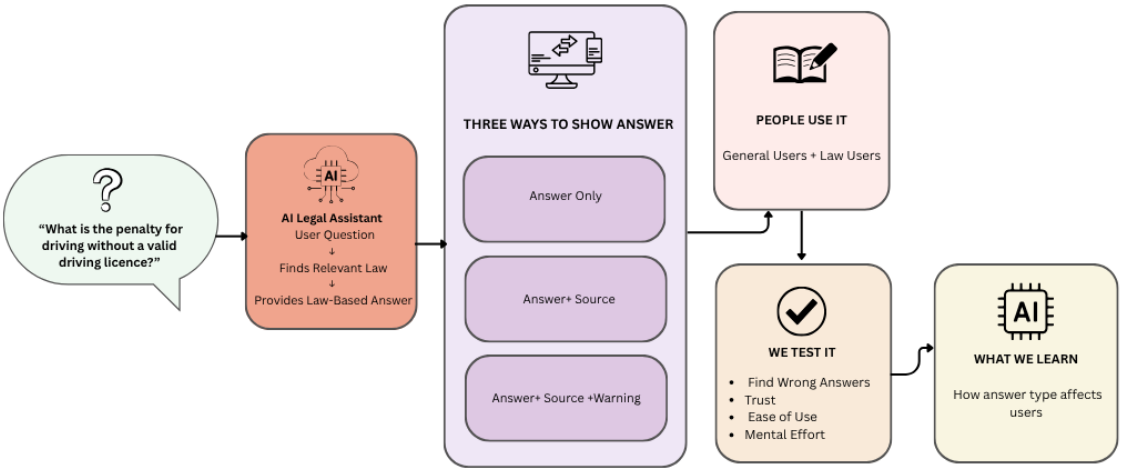



Figure 1.1: Graphical abstract of the proposed explainable legal assistant.

### 1.4.1 Statutory Data Collection and Dataset Creation

The *Bangladesh Road Transport Act, 2018* serves as the primary legal source for this study. The document is processed through text extraction, cleaning, and semantic segmentation while preserving its original legal structure. Each segment contains metadata such as section number, clause number, page number, and source information to support retrieval and citation generation.

### 1.4.2 Validated Legal Question–Answer Dataset

A Bangla legal question-answer dataset is created from the *Bangladesh Road Transport Act, 2018*. The dataset includes factual questions, procedural queries, scenario-based problems, and edge cases representing common legal information needs. Each question is linked with relevant legal provisions and manually validated for accuracy and consistency.

### 1.4.3 Development of the RAG-Based Legal Assistant

The proposed Legal AI system uses a Retrieval-Augmented Generation (RAG) architecture to generate source-grounded responses. Legal documents are converted into embeddings and stored in a vector database. For each user query, the system retrieves relevant legal sections and provides them as context to an open-weight Large Language Model (LLM), such as Llama, Qwen, or Gemma. The system then generates answers with relevant legal references to improve reliability and transparency.

### 1.4.4 Development of Experimental User Interfaces

Three web-based interfaces are developed using the same RAG pipeline to study the effect of explainability features on user trust and hallucination detection.

- **Treatment A (Baseline Interface):** Provides responses through a simple chatbot without citations or explanations.
- **Treatment B (Traceability Interface):** Adds clickable citations that show relevant sections, clauses, pages, and source text from the *Bangladesh Road Transport Act, 2018*.
- **Treatment C (Explainable AI Interface):** Adds confidence indicators, uncertainty highlights, and warning messages to help users identify potentially unreliable responses.

All interfaces use the same AI model and retrieval process. This allows the study to measure the effect of interface design on user trust, hallucination detection, and response evaluation.

## 1.5 Project Outcome

The expected outcomes of this project are:

1. Development of a RAG-based Legal AI question-answering system for Bangladeshi Traffic Law.
2. Creation of an expert-validated Bangla legal dataset containing legal scenarios, question-answer pairs, and statutory references.
3. Design and evaluation of multiple Human–AI interfaces with different levels of explainability, source traceability, and uncertainty visualization.
4. Analysis of how interface design and user expertise affect hallucination detection, trust calibration, cognitive workload, and usability.
5. Development of design recommendations for building transparent and trustworthy Legal AI systems for Bangla-speaking users.
6. Contribution to research in Human–Computer Interaction (HCI), Explainable Artificial Intelligence (XAI), and Legal Technology in low-resource language settings.

## 1.6 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 1** introduces the research background, LLM hallucination challenges in the legal domain, and the proposed RAG-based Legal AI approach.
- **Chapter 2** discusses the motivation, research context, and technical and ethical considerations behind developing a Legal AI system for the *Bangladesh Road Transport Act, 2018*.

- **Chapter 3** presents the project objectives, including dataset development, RAG system design, model evaluation, interface design, and user trust analysis.
- **Chapter 4** describes the research methodology, including data collection, system implementation, and interface evaluation.
- **Chapter 5** summarizes the expected outcomes, contributions, evaluation findings, and future directions for trustworthy Legal AI systems.

# Chapter 2

# Background

This chapter reviews the main technical and human-centred concepts related to the proposed legal-assistance system, as well as recent studies on Retrieval-Augmented Generation (RAG) in Bangladeshi law. It also highlights a key research gap: the limited understanding of how source traceability and uncertainty-focused interface design can help users identify legal-AI errors and decide how much to rely on RAG-generated responses.

## 2.1 Preliminaries

In this section, we provide the background knowledge necessary to understand the rest of the report. The main ideas include the following.

### 2.1.1 Large Language Models and Legal Hallucination

Large Language Models (LLMs) are AI models which can generate text by predicting likely words based on patterns learned from large amounts of data. They are not always accurate, although their responses can sound natural and convincing.

A *hallucination* occurs when an LLM produces false information as though it is correct. In the legal sector, this can include inventing laws or sections, giving incorrect penalties or deadlines, or presenting outdated legal information.

### 2.1.2 Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) combines an LLM with information retrieved from an external collection of documents.

In legal works, RAG helps the model to base the response on relevant legal sources, such as Acts and sections, instead of relying only on trained data.

### 2.1.3 RAG Variants Relevant to Bangladeshi Law

A *vanilla RAG* system retrieves relevant information once and then uses it to generate an answer. More advanced RAG systems add more steps: such as query rewriting, relevance

checking, multi-stage retrieval, hybrid search, and re-ranking to improve the quality of the retrieved information.

### 2.1.4 Bangla and Jurisdiction-Specific Constraints

Legal AI in Bangladesh must handle both the language and jurisdiction-specific challenges. Legal documents may appear in Bangla, English, or a mixture of both, while spelling differences, transliteration, formal legal language, and OCR errors can make the retrieval more difficult than usual.

The system must also retrieve the correct law, section, and the version that applies to the user's question. A response may sound convincing, but still be legally incorrect if it relies on an irrelevant or outdated information.

### 2.1.5 Explainability, Traceability, and Trust Calibration

In this project, *explainability* refers to the information which helps the user to understand the basis of an AI-generated answer, such as citations, supporting legal text, and details of the source.

*Traceability* refers to the ability to connect a generated answer to the supported legal source.

*Trust calibration* means how well a user's level of trust match with the actual correctness of the system's response.

### 2.1.6 Technical and Human-Centred Evaluation

*Technical evaluation* measures how good the system performs in terms of retrieval quality, correctness of the answer, faithfulness, robustness, and efficiency.

*Human-centred evaluation* focuses on the user experience when interacting with the system. It considers factors such as finding errors, checking evidences, confidence, task completion time, mental effort, and overall usability.

## 2.2 Literature Review

### 2.2.1 Similar Applications

#### JuristRAG: An Enhanced RAG System for Semantic Question Answering on Legal Statutes (2025)

JuristRAG is a system based for legal question-answering based on the Code of Criminal Procedure, the Penal Code, and the Special Powers Act of Bangladesh[5]. The study shows that advanced retrieval strategies improve answer quality compared to a vanilla RAG pipeline. However, the evaluation mainly focuses on system performance metrics and expert assessment, with limited investigation of user-centred factors such as trust,

confidence, usability, cognitive effort, and reliance on AI-generated legal advice. This highlights an opportunity for further research from an HCI perspective.

**LegalRAG: A Hybrid RAG System for Multilingual Legal Information Retrieval (2025)**

LegalRAG aims for the bilingual Bangladesh Police Gazettes. It is especially relevant to low-resource and noisy regulatory documents [6]. The system utilizes multilingual embeddings, a vector database, and Maximum Marginal Relevance retrieval. Its advanced pipeline introduces an extra LLM for relevance checking and query refinement before final generation. Evaluation on a curated 168-question dataset and three human evaluators shows improvements over vanilla RAG for multiple generators. Above all, the paper also reports a persistent weakness on out-of-context questions, where both vanilla and advanced pipelines may give incorrect answers with high confidence. This result is directly connected to safety-oriented interface design because it shows that a technically improved retriever does not automatically guarantee a safe response for the end-user.

**JusticeNetBD: Context-Aware AI to Enhance Legal Information Access for Bangladeshi Women via Retrieval-Augmented Generation (2025)**

JusticeNetBD is a RAG-based legal assistant designed around legal information access for Bangladeshi women [7]. Its corpus combines statutory provisions with institutional and legal-aid information, and its publicly accessible prototype returns retrieved evidence together with generated answers. The work reports strong retrieval metrics and low response latency compared with several closed-book commercial LLMs. It therefore demonstrates the practical value of a lightweight, domain-grounded assistant. Nevertheless, the quantitative benchmark is small, and the claimed accessibility and culturally sensitive design are not validated through a controlled study with the intended end-user population. The paper itself identifies broader corpus coverage, colloquial-query handling, and multilingual support as future work.

**Mina: A Multilingual LLM-Powered Legal Assistant Agent for Bangladesh for Empowering Access to Justice (2025)**

Mina is the broadest of the reviewed Bangladeshi legal assistants [8]. It combines multilingual embeddings, an orchestrating agent, two-stage Act–Section retrieval, translation, legal-dictionary support, document generation, citations, and selective tool use. The system is evaluated across Bangladesh Bar Council examinations and shows substantial improvements when moving from no RAG to naïve RAG, two-step RAG, and tool-augmented configurations. Its error analysis is especially informative: retrieval improves performance, yet hierarchical procedural reasoning, timeline calculations, fact-sensitive adaptation, and clarification remain difficult. Mina also discusses confidence indicators, provenance interfaces, clarification, and human review as safeguards. However, its reported evaluation is primarily an expert/examination evaluation rather than a controlled comparison of user-interface treatments across lay and legally trained users.

**A Novel Evaluation Strategy for Domain-Specific Legal QA: Application to Bangladeshi Law with RAG (2026)**

LawRAG develops a Bangladeshi legal QA pipeline together with a domain-specific evaluation framework [9]. Its retrieval architecture combines semantic search, TF–IDF lexical search, intent-aware query reformulation, acronym expansion, and cross-encoder reranking. The work proposes a Composite Legal Reliability score integrating conventional semantic evaluation with numerical accuracy, temporal validity, LLM-based judgment, statute validity, and robustness. This is a useful contribution because it makes explicit that a legally fluent answer may still fail on dates, penalties, or statutory validity. At the same time, the evaluation is explicitly pilot-scale, the system is mainly English-language, updates require manual handling, and complex legal reasoning remains difficult. The work also identifies multilingual support, human-in-the-loop verification, and greater explainability as future directions.

**Other recent work:**

The existing literature suggests that several aspects of Bangladeshi Legal AI have already been explored. For example, a multi-agent RAG system has introduced a clarification agent for legal information retrieval [10], and recent evaluation of LLMs on 250 authentic Bangladeshi legal questions has shown that realistic user queries and expert evaluation are possible, while fluent responses may still contain harmful misinformation [11]. Therefore, this project does not claim novelty in areas such as realistic legal questions, multilingual support, citations, retrieval-based systems, or explainability. Instead, its contribution focuses on evaluating how these systems affect real users' understanding and trust.

Recent legal AI benchmarks also highlight the need for domain-specific and user-focused evaluation. LegalBench evaluates 162 expert-designed legal reasoning tasks, while LawBench shows that LLMs still struggle with many legal tasks. LLeQA further demonstrates that strong automatic evaluation scores do not always guarantee factual accuracy in long-form legal answers [12, 13, 14]. ARES addresses this challenge by evaluating RAG systems through context relevance, answer faithfulness, and answer relevance while reducing manual evaluation effort [15].

Recent studies also reveal specific reliability issues in legal AI systems. CLAW highlights challenges in statute recall and historical legal versions, while LEGIT and ATRIE show that retrieval-based approaches can improve legal reasoning but still leave accuracy gaps compared to expert responses [16, 17, 18]. Other works, such as AALawyer and PubLawBench, further show that reducing hallucinations and achieving professional-level legal performance do not automatically ensure safe and understandable assistance for ordinary users [19, 20].

### 2.2.2 Related Research

#### Methodological Patterns

Across the reviewed Bangladeshi systems, RAG is treated primarily as an information-retrieval and grounding problem. JuristRAG and LegalRAG refine queries when retrieval is weak; Mina adds hierarchical Act–Section retrieval and tools; LawRAG fuses semantic and lexical retrieval with reranking; and JusticeNetBD uses a lightweight dense-retrieval pipeline for fast evidence-backed responses. The shared assumption is that better retrieved context should produce safer or more accurate answers. The empirical results generally support the first part of this assumption: advanced retrieval commonly improves automatic or expert scores.

However, the literature also shows the limits of this assumption. LegalRAG still fails on out-of-context queries, Mina finds that retrieval alone does not solve hierarchical and procedural reasoning, and LawRAG reports difficulty with complex reasoning and numerical legal constraints [6, 8, 9]. Recent broader legal-RAG evaluation likewise shows that hallucinations remain present after retrieval augmentation [21]. Thus, improved retrieval is necessary but not sufficient for safe legal assistance.

#### Dataset and Validation Patterns

The datasets are heterogeneous but predominantly system-oriented. JuristRAG uses three major criminal statutes and an automatically curated 700-pair QA evaluation set. LegalRAG uses 13 Police Gazettes and a 168-pair dataset generated with an LLM and subsequently checked by authors. JusticeNetBD reports a 10-question benchmark over a women-focused legal corpus. LawRAG indexes a much larger corpus but evaluates its legal-reliability framework on 20 pilot questions. Mina uses the 2022 and 2023 Bangladesh Bar Council examinations, giving it stronger professional-task coverage but a population and task distribution that differs from ordinary citizen legal help.

This pattern limits external validity for consumer-facing legal assistance. It does not mean that authentic Bangladeshi legal-question datasets do not exist: recent work has evaluated models on 250 authentic questions gathered from a public legal-help community [11]. The narrower limitation is that the reviewed RAG applications have not yet shown that their technical gains transfer to how ordinary users interpret, verify, and act on answers in realistic interactions.

#### Application and Human-Factors Patterns

Existing Bangladeshi Legal AI applications target different groups such as criminal-law users, women who seek legal information, communities with low-income, bilingual users, and legal professionals. This shows progress in localizing Legal AI systems.

However, the impact of interface design on users with different levels of legal expertise is still underexplored. Previous work, such as *ContractMind*, shows that the interface

design can influence trust calibration in legal AI systems [22]. A key research gap remains in understanding whether features like source traceability and uncertainty indicators can improve users' ability to evaluate Bangla legal AI responses, especially across both legal and non-legal users.

#### Temporal Pattern

Earlier legal-AI work raised concerns about hallucination and verification, and recent research shows that they have not been convincingly eliminated. Commercial legal research systems have been observed to hallucinate despite retrieval augmentation [23], while a 2026 study across multiple legal RAG systems reports continuing claim-level hallucination and particular vulnerability to false-premise questions [21]. The technical problem is therefore persistent. The research opportunity for this project is not to promise an impossible zero-error model, but to examine how the interface can help users recognize and respond appropriately when residual errors occur.

## 2.3 Gap Analysis

The reviewed literature shows substantial progress in the technical development of legal RAG systems for Bangladesh. Existing work has already explored iterative retrieval, query decomposition, multilingual retrieval, relevance checking, two-stage Act–Section retrieval, hybrid dense–lexical search, legal-specific evaluation, citation generation, and practical legal-assistance applications [5, 6, 7, 8, 9]. Therefore, the research gap for this project should not be framed as the absence of Bangladeshi legal RAG, multilingual legal assistance, or source-backed legal question answering. Instead, the gap emerges from how these systems are evaluated and how their remaining errors are presented to end users.

Table 2.2 summarizes the main gaps identified from the literature and distinguishes between areas that are already substantially addressed and areas that remain suitable for further investigation.

The strongest gap for the proposed project is therefore a *socio-technical and human-centred evaluation gap*. Existing Bangladeshi legal RAG research primarily asks whether a retrieval or generation strategy improves system performance. In contrast, comparatively little evidence establishes what happens after an imperfect answer is shown to a user. A RAG system may retrieve the wrong provision, omit an exception, misstate a penalty, apply a correct section to the wrong factual situation, or answer an out-of-scope question with unwarranted confidence. Even when the interface displays a citation or retrieved passage, the literature does not establish whether users notice the evidence, understand it, verify the answer against it, or change their reliance appropriately.

This distinction is important because the goal of a high-stakes legal assistant should not be to maximize user trust. The appropriate objective is *calibrated reliance*: users should be more willing to accept answers that are well supported and more willing to question or

Table 2.1: Gap analysis of existing Bangladeshi legal RAG research

| **Research Dimension** | **Evidence from Existing Work** | **Gap Status** | **Implication for This FYDP** |
|---|---|---|---|
| Bangladeshi legal RAG development | JuristRAG, LegalRAG, JusticeNetBD, Mina, and LawRAG already demonstrate jurisdiction-specific RAG systems for Bangladeshi statutes and regulatory documents [5, 6, 7, 8, 9]. | **Not a gap** | The project should not claim novelty simply from applying RAG to Bangladeshi law. |
| Bangla and multilingual support | LegalRAG and Mina already use multilingual retrieval for Bangla and English, while other systems identify multilingual support as an extension [5, 6, 8]. | **Partially addressed** | Native-Bangla support remains useful for the project, but it is not sufficient as the primary research contribution. |
| Source traceability and explainability | Existing systems expose retrieved context, generate citations, or propose provenance and confidence-oriented safeguards [5, 7, 8]. However, their effectiveness is mainly discussed as a system feature rather than tested through controlled behavioral comparisons. | **Synthesized gap** | Compare alternative presentation strategies to determine whether users can actually use evidence cues to detect incorrect answers. |
| Human trust calibration and error detection | Most Bangladeshi studies emphasize retrieval, generation, expert correctness, similarity, or legal-reliability metrics rather than measuring whether end users correctly accept or reject individual AI answers [5, 6, 7, 9]. | **Strong synthesized gap** | Make hallucination detection and calibrated reliance primary behavioral outcomes rather than treating user trust as a general satisfaction measure. |
| Effect of legal expertise | Existing evaluations largely use legal experts, law students, faculty, or benchmark-style questions, while intended public-facing systems also target non-experts [5, 6, 8]. | **Synthesized gap** | Explicitly compare lay users and legally trained users to test whether expertise changes the usefulness of transparency cues. |
| Out-of-scope, ambiguous, and false-premise queries | LegalRAG reports high-confidence errors on out-of-context questions, while JuristRAG and Mina identify the need for better clarification, confidence handling, and safe responses [5, 6, 8]. | **Explicit/secondary gap** | Use such queries as controlled failure cases and evaluate whether the interface helps users recognize when an answer should not be trusted. |
| Relationship between technical metrics and human safety | The literature reports faithfulness, retrieval accuracy, semantic similarity, legal-specific reliability, and expert scores, but does not establish whether these values predict user-level detection or reliance [5, 7, 9]. | **Synthesized secondary gap** | Log answer-level technical scores and test whether they predict correct human reliance decisions. |

Table 2.2: Gap Analysis of Existing Legal AI and Retrieval-Augmented Generation Systems

| **Model / Paper** | **BD Traffic Law** | **Bangla–English** | **Legal Structure Retrieval** | **Hybrid Retrieval** | **Query Refinement / Reranking** | **Temporal Awareness** | **Traceable Legal Source** | **Citation Verification** | **Retrieval Evaluation** | **Legal Expert Evaluation** |
|---|---|---|---|---|---|---|---|---|---|---|
| Mina | ✗ | ✓ | ✓ | △ | ✓ | ✗ | ✓ | ✗ | ✓ | ✓ |
| JuristRAG | ✗ | ✗ | ✓ | ✓ | ✓ | ✗ | △ | ✗ | ✓ | ✓ |
| JusticeNetBD | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | △ | ✗ | ✓ | ✗ |
| LegalRAG | ✗ | ✓ | ✗ | ✗ | ✓ | △ | △ | ✗ | ✗ | ✓ |
| LawRAG | ✗ | ✗ | △ | ✓ | ✓ | ✓ | △ | △ | ✓ | ✓ |
| A Visual Search Engine for BD Laws | ✗ | △ | ✓ | ✓ | ✗ | ✓ | ✓ | △ | ✗ | ✓ |
| Context-Injected Fine-Tuning for BD Law | ✗ | ✓ | △ | ✗ | ✗ | ✗ | ✓ | ✗ | ✗ | ✗ |
| Pathways to Smart Justice | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | △ | ✗ | ✗ | ✓ |
| Reliability of LLMs in Bengali Legal Context (Aftahee et al., 2025) | ✗ | ✓ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| Efficient LLM Distillation for Bangladesh Legal | ✓ | △ | △ | ✓ | ✓ | ✗ | ✓ | ✗ | ✗ | ✓ |
| Legalyze – An Open-Source Structure-Aware Agentic RAG Platform | ✗ | ✗ | ✓ | ✓ | ✓ | ✗ | ✓ | ✗ | ✓ | ✓ |
| Exploring Possibilities of AI-Powered Legal Assistance | ✓ | ✗ | ✗ | ✗ | ✗ | △ | △ | ✗ | ✗ | ✓ |
| **Proposed Model** | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

**Legend:** ✓ = Supported; △ = Partial / Planned; ✗ = Not supported.

reject answers that are incorrect, weakly supported, ambiguous, or outside the corpus. A system can therefore achieve strong average retrieval or semantic-similarity scores while still creating risk if users cannot identify the remaining failures. The persistent out-of-context errors reported by LegalRAG and the procedural and fact-sensitive reasoning weaknesses reported by Mina reinforce this concern [6, 8]. LawRAG similarly demonstrates that legal correctness requires dimensions such as temporal validity, numerical accuracy, statutory validity, and robustness that are not captured by generic NLP similarity alone [9].

The proposed FYDP addresses this gap by holding the underlying legal-RAG task approximately constant while experimentally varying the way system output is presented. The initial corpus will be based on the Bangladesh Road Transport Act, 2018, and the study will compare three interface conditions: a plain conversational baseline, a traceability-oriented interface with direct access to supporting statutory text, and an uncertainty-oriented interface that adds warnings or visual cues when system support is weak. These interface conditions will be evaluated across two user groups: participants without formal legal training and participants with legal-domain knowledge.

The central contribution is therefore not the introduction of another RAG retrieval algorithm. Rather, the project investigates whether different transparency mechanisms improve a user's ability to detect legal-AI errors and whether the effect depends on legal expertise. The primary outcome measures should be behavioral: hallucination-detection rate, acceptance of incorrect answers, rejection of correct answers, task accuracy, verification behavior, and decision confidence. Cognitive load and usability can be measured using NASA-TLX and the System Usability Scale (SUS), allowing the study to determine whether additional transparency improves safety at an acceptable interaction cost.

Based on this analysis, the primary research question is formulated as follows:

> *How do source-traceability and uncertainty-oriented interface designs affect hallucination detection, calibrated reliance, cognitive load, and usability among Bangladeshi users with different levels of legal expertise when interacting with a RAG-based legal assistant?*

Two secondary questions extend the same research direction:

> *How does legal-domain expertise moderate the effect of interface transparency on users' ability to distinguish correct from incorrect legal-AI answers?*
>
> *Which answer-level technical signals, such as retrieval relevance, source support, or legal-reliability measures, are associated with correct human reliance decisions?*

A further secondary engineering question concerns safe failure:

> *How should a legal RAG assistant communicate uncertainty or lack of sufficient evidence when a query is ambiguous, false-premised, or outside the coverage of its legal corpus?*

The expected contribution is a controlled evaluation framework for human interaction with Bangladeshi legal RAG systems. Specifically, the project can contribute: (i) an expert-validated native-Bangla statutory QA and controlled-error test set for the selected legal domain; (ii) an experimental comparison of baseline, source-traceability, and uncertainty-oriented interfaces; (iii) empirical evidence on whether legal expertise moderates error detection and reliance; (iv) analysis of the relationship between backend reliability indicators and user behavior; and (v) design recommendations for safer public-facing legal AI systems in Bangladesh. This framing is deliberately conservative: it does not claim that explainability, citations, multilingual RAG, or trust research are themselves new. The contribution lies in experimentally evaluating how these mechanisms affect behavioral safety in the specific context of Bangladeshi legal RAG.

## 2.4 Summary

This chapter reviewed the technical foundations of legal RAG and five closely related Bangladeshi systems. The literature shows clear progress in multilingual retrieval, query refinement, legal-specific evaluation, agentic workflows, and practical deployment, but also shows persistent failures involving out-of-context queries, complex legal reasoning, numerical and temporal constraints, and residual hallucination. Most importantly for this FYDP, existing Bangladeshi work evaluates the quality of the *system* more extensively than the safety of the *human–AI interaction*. The project therefore focuses on a deliberately scoped research gap: experimentally determining whether traceability and uncertainty-oriented interface designs help lay and legally trained Bangladeshi users detect errors and calibrate reliance without imposing unacceptable cognitive or usability costs.

# Chapter 3

# Project Design

This chapter covers the system requirements, project plan, and division of work for the proposed explainable legal assistant. It includes a RAG-based legal question-answering backend coupled with controlled interface treatments for studying hallucination detection and user trust.

## 3.1 Requirement Analysis

The project has two interconnected components. The first is a legal question-answering backend that retrieves relevant provisions from an authoritative Bangladesh traffic-law corpus and utilizes a large language model to generate a grounded response. The second is a web interface that presents the same underlying legal-assistance task at varying levels of source traceability and uncertainty information. The initial legal scope is the Bangladesh Road Transport Act, 2018 though the design is amenable to later corpus expansion.

### 3.1.1 Functional and Nonfunctional Requirements

**Functional requirements**

1. The system will store the traffic laws of Bangladesh in an organized database. The data would be extracted from official and trustworthy sources.
2. The system will redeem crucial specifics for each law. This will include the Act name, section number, subsection or clause and the source.
3. End-users can inquire with a query in normal language. The primary tongue will be Bangla.
4. The system would search in the data base for relevant legal information. It would retrieve the sections that best suits the user's question.
5. Then the system would use the retrieved information to formulate an answer. It would base the replies on the relevant traffic laws.

6. The web application shall support three experimental interface conditions: a plain baseline interface, a traceability interface with source access, and an explainable interface with additional uncertainty or evidence-support cues.
7. The traceability and explainable interfaces shall allow the user to inspect the legal source connected to an answer or claim.
8. The explainable interface shall present a warning or other visible cue when the available backend evidence is weak or insufficient according to the selected support criterion.
9. The research prototype shall record the interaction data required for evaluation, such as the selected interface condition, user decisions, source-verification actions, and task responses.
10. The evaluation workflow shall support both users without formal legal training and users with legal-domain knowledge.

**Nonfunctional requirements**

1. **Legal grounding:**The system should use relevant Bangladeshi traffic laws as the source of information when answering legal questions.
2. **Traceability:** Users should be able to see the source or section of the laws that has been used to support the answers.
3. **Usability:** The interface should be easy for general users to understand without formal legal training or knowledge of the underlying RAG architecture.
4. **Performance:** The system should retrieve information and generate answers within a reasonable time for a web-based application.
5. **Consistency:** All the different interfaces should use the same backend logic and equivalent task content so that interface effects are not mixed with backend changes.
6. **Privacy and ethics:** The study should collect only the information needed for analysis, protect participants privacy, store research data securely, and obtain consent from users before testing.
7. **Maintainability:** All the components, for example, data processing, retrieval, generation, interface, and evaluation, should remain modular so that one component can be changed without affecting the others or rewriting the full system.
8. **Reproducibility:** All the procedures, for example, model settings, retrieval settings, study conditions, and evaluation procedures, should be recorded so that the experiments can be repeated.
9. **Extensibility:** The system should be designed so that more Bangladeshi laws or other legal areas can be added in the future.

### 3.1.2 Context Diagram

The context diagram defines the boundary of the proposed system and the external entities that exchange information with it. A user or study participant submits a legal question and receives an AI-generated answer along with the information shown by the assigned interface. Official legal documents will be used as the main source for the legal dataset. Legal validators or research team members check the question-answer dataset and the legal sources used in the experiment. The research team also sets up the study and collects the resulting interaction and evaluation data.

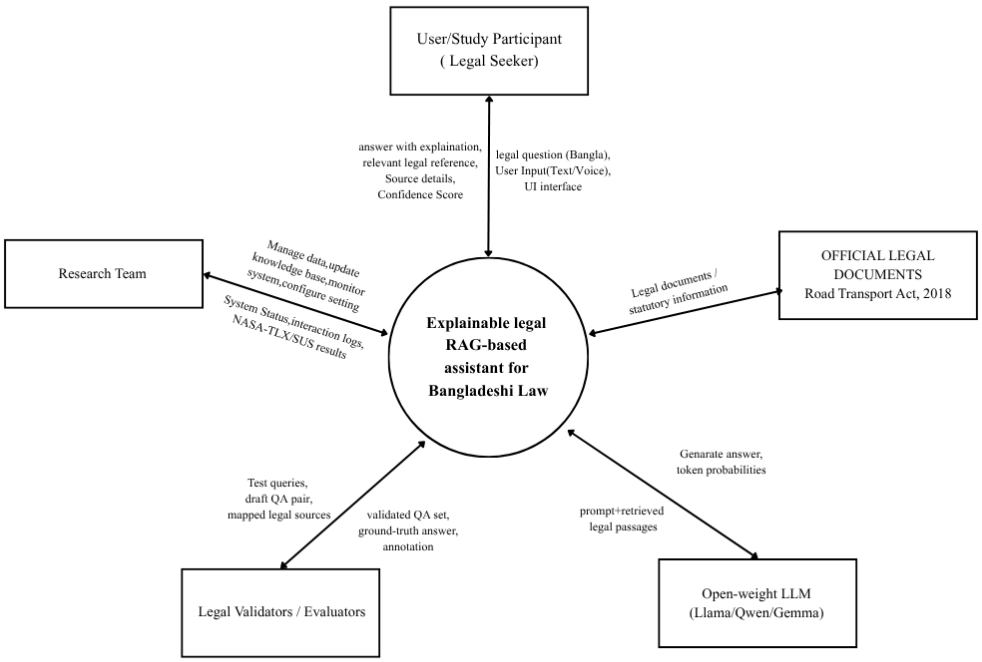



Figure 3.1: Context diagram of the proposed explainable legal assistant.

### 3.1.3 Data Flow Diagram Level 1

The Level 1 Data Flow Diagram (DFD) can be divided into its processes. Firstly, legal documents are prepared and stored in the repository of legal knowledge. Secondly, after an entry is received, the query-processing and retrieval process searches the repository for applicable legal passages. The grounded answer-generation process uses these passages to formulate a response with sources. The presentation process then renders the resulting answer according to the chosen interface condition. Finally, the study records the user's decisions and interaction events, which can be used for further evaluation purposes. Therefore, the DFD's data stores include the legal knowledge base, the validated question-answer or study data set, and the interaction and evaluation records.

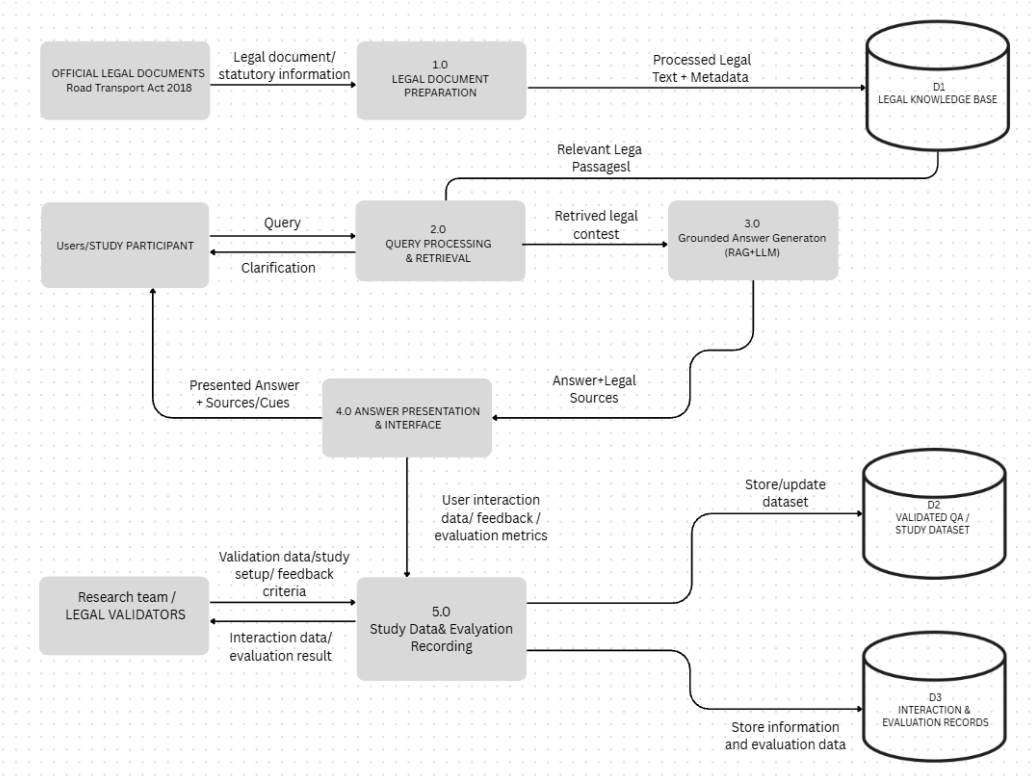



Figure 3.2: Level 1 data flow diagram of the proposed system.

### 3.1.4 UI Design

The web application will use a common conversational layout so that the experimental conditions differ mainly in the information shown with the legal answer. The first condition is a baseline chatbot interface that presents the generated answer without source-tracking or uncertainty cues. The second condition adds source traceability through inline citations or source controls that allow the user to inspect the relevant legal provision. The third condition retains the traceability features and adds a limited set of uncertainty or evidence-support cues, including a visible warning when the backend evidence is considered insufficient.

These interfaces are experimental conditions rather than separate products. The baseline provides a control condition, the traceability interface tests the effect of direct source access, and the explainable interface tests whether an additional uncertainty cue changes error detection, reliance, workload, or usability. The final interface design will use the same terminology, navigation pattern, and general visual structure across all three conditions to reduce unrelated UI differences.

## 3.2 Detailed Methodology and Design

## 3.3 Project Plan

The project will be completed in phases throughout the one-year FYDP period. Certain activities are overlapping, although each phase has a clear dependency and a milestone to indicate that the team has finished enough work to proceed to the next one.

Table 3.1: Project phases, dependencies, and milestones.

| **Phase** | **Main Activities** | **Dependency** | **Milestone** |
|---|---|---|---|
| 1. Requirement gathering and study design | Finalize scope, requirements, research questions, and evaluation plan. | None | M1: Scope and requirements approved. |
| 2. Corpus and dataset preparation | Collect and structure traffic-law material and prepare the validated Bangla QA/study set. | M1 | M2: Corpus and initial dataset ready. |
| 3. RAG backend development | Implement indexing, retrieval, grounded generation, and source tracking. | M2 | M3: End-to-end legal QA prototype working. |
| 4. Interface and integration | Build the baseline, traceability, and explainable interfaces and connect them to the backend. | M3 | M4: Integrated experimental prototype ready. |
| 5. Pilot and refinement | Test study tasks, logging, interface clarity, and backend behavior; fix identified problems. | M4 | M5: Pilot completed and protocol finalized. |
| 6. Main evaluation | Run technical evaluation and the user study with lay and legal-domain participants. | M5 | M6: Evaluation data collected. |
| 7. Analysis and reporting | Analyze results, document limitations, and prepare the final thesis and presentation. | M6 | M7: Final thesis and project deliverables completed. |

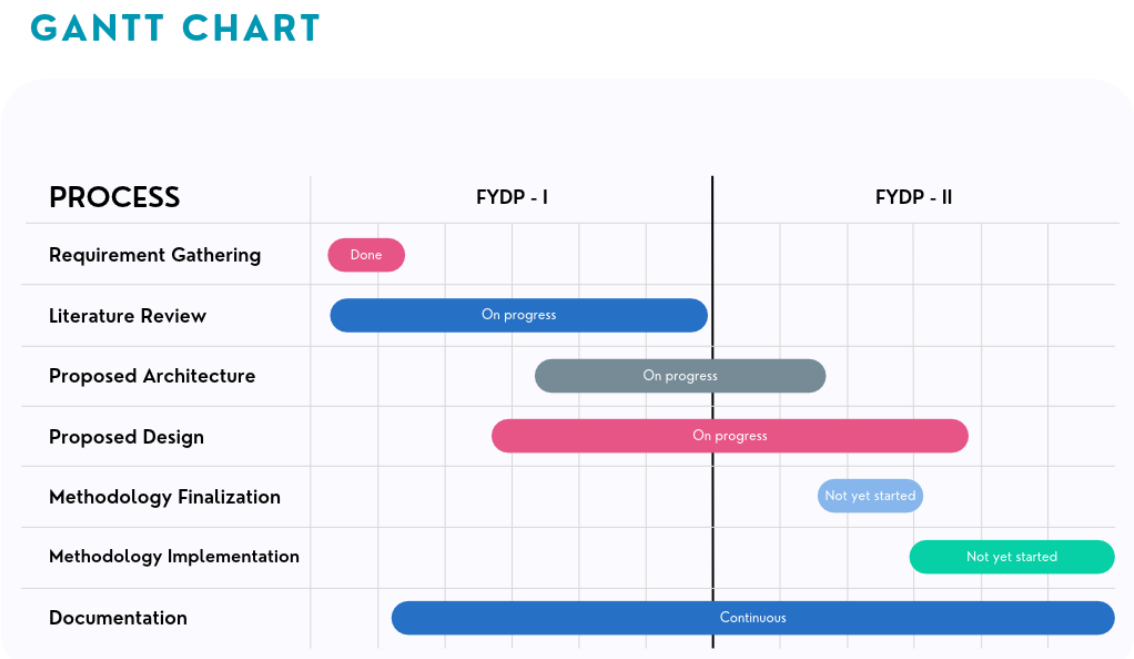



Figure 3.3: Gantt chart of the Project

### 3.3.1 SWOT Analysis

The SWOT analysis summarizes internal project characteristics and external conditions that may affect implementation and evaluation.

Table 3.2: SWOT analysis of the proposed project.

| | |
|---|---|
| **Strengths**<br>• Uses official and reliable Bangladeshi legal documents.<br>• Uses RAG to provide answers based on legal sources and keeps source information.<br>• Includes an expert-validated Bangla QA/study dataset.<br>• Combines technical evaluation with a controlled user study. | **Weaknesses**<br>• Initial coverage is limited to a small legal domain.<br>• RAG cannot guarantee that all hallucinations will disappear.<br>• Bangla retrieval quality depends on available models and data.<br>• Recruiting enough legal-domain participants may be difficult. |
| **Opportunities**<br>• Extend the System to other Bangladeshi laws.<br>• Produce a reusable Bangla legal QA/evaluation resource.<br>• Study safer source-traceability and uncertainty interfaces.<br>• Add stronger citation verification or abstention in the future. | **Threats**<br>• Changes in laws may make stored legal information outdated.<br>• Model or API availability and cost may change during the project.<br>• Users may over-trust imperfect confidence or warning cues.<br>• Time limits may restrict the number of models and study cases that can be tested. |

## 3.4 Task Allocation

Each member will have primary ownership of one major module while taking part in integration, testing, documentation, and presentation work. The allocation is intended to keep the main technical and research responsibilities comparable across the five members.

Table 3.3: Module ownership across the five team members.

| **Team Member** | **Primary Module** | **Main Responsibilities and Deliverables** |
|---|---|---|
| Sifatullah | Legal corpus and knowledge-base preparation | Collect and verify authoritative legal documents; clean and structure statutory text; preserve section-level metadata; maintain the legal knowledge base; support corpus quality checks and later updates. |
| Tasnuba Haque | QA dataset and retrieval pipeline | Prepare the Bangla QA/benchmark workflow; coordinate question-to-clause mapping; implement embeddings, indexing, retrieval, and related retrieval tests; document retrieval settings and failure cases. |
| Ibtida Ara | LLM generation and reliability layer | Design grounded prompts; integrate the selected LLM; generate source-linked answers; implement answer-support, warning, or citation checks required by the prototype; run generation-quality tests. |
| Aminul Islam Rifat | Frontend and system integration | Develop the common web interface and the three experimental UI conditions; connect the frontend to backend services; implement source inspection and study-event logging; support end-to-end integration testing. |
| Ifrat-E-Sarwar Orpa | User study and evaluation | Prepare the study protocol and participant materials; coordinate pilot and main user testing; manage NASA-TLX, SUS, trust, and hallucination-detection measurements; organize and analyze evaluation data. |

All five members will contribute to literature review updates, requirement reviews, dataset checking, integration tests, interpretation of results, thesis writing, and presentation preparation. Cross-review of another member's module will be used before each major milestone so that no critical component depends on a single person.

## 3.5 Summary

# Chapter 4

# Implementation and Results

[Must be present in Final Report. Incomplete version might be included in FYDP-1 Report, however it is optional.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

## 4.1 Environment Setup

## 4.2 Testing and Evaluation

## 4.3 Results and Discussion

## 4.4 Summary

# Chapter 5

# Standards and Design Constraints

This chapter focuses on the software, hardware, and communication standards followed during the development of the Explainable Legal Assistant for Bangladeshi Law. It also explores our design constraints, provides a detailed cost analysis of the project, and outlines how our system maps to complex engineering problem-solving (P1-P7) and complex engineering activity (A1-A5) criteria specified by the Board Accreditation for Engineering and Technical Education (BAETE).

## 5.1 Compliance with the Standards

To ensure reliability, scalability, and long-term maintainability of our legal assistant system, we guided our design choices through strict adherence to key industry standards throughout the development cycle.

### 5.1.1 Software Standards

To ensure quality and reproducibility, the software architecture integrates multiple important standards.

**Programming and Development Standards:** For our software development, we adhered to Python Enhancement Proposal (PEP) 8 standards for Python programming to ensure readability and maintainability of the codebase.

**RAG Framework and Model Selection:** We utilized [LangChain / LlamaIndex] for RAG orchestration and the sentence-transformers ecosystem with [embedding model, e.g., multilingual-e5] to embed the Road Transport Act, 2018 corpus. Embeddings are stored in [FAISS / ChromaDB] to retrieve relevant statutory sections. Answer generation is performed by interfacing with open-weight Large Language Models (LLMs) such as Llama, Qwen, or Gemma.

**Alternate Options:** We considered using closed commercial APIs (such as GPT or Gemini) for generation and building a fully custom retrieval pipeline from first principles, without relying on any existing RAG framework or open-weight model.

**Rationale for Selection:** Existing RAG libraries and the sentence-transformers ecosystem allow rapid prototyping and easy integration of pretrained multilingual embedding models with open-weight LLMs. Open-weight models were preferred over closed APIs for three reasons. First, they give direct access to token-level probabilities, which the confidence indicators in Treatment C require, whereas closed APIs offer limited or inconsistent access to them. Second, they keep the experiment reproducible, since model versions do not change silently on the provider's side. Third, they avoid per-query API costs during the user study. The citation-traceability needed for the Treatment B does not depend on this choice, because it comes from the retrieval metadata (section, clause, page) of our own pipeline. A fully custom pipeline would offer more control, but it would take much longer to build and add little value, since the research contribution lies in the interface design and not in the retrieval algorithm.

**Text Encoding:** All statutory texts, questions, and answers are handled in UF-8 (Unicode) to correctly store, retrieve, and display Bangla script. Legacy non-Unicode Bangla encodings were avoided because they break retrieval and rendering.

**Data Formats and API:** Chunk metadata (section, clause, page number) and the question-answer dataset are stored in JSON and exported in CSV for analysis.The frontend communicates with the RAG backend through a REST API using JSON.

**Web and Accessibility:** The three experimental interfaces are built with HTML5, CSS3, and JavaScript and follow WCAG 2.1 (Level AA) guidelines for contrast, font size and keyboard navigation, since participants include general users with no legal or technical background.

**Human-Centered Design and Evaluation:** The interface design follows SO 9241-210 (Human-Centered Design).User cognitive workload and usability are measured using the standard NASA-TLX and System Usability Scale (SUS) instruments.

**Security:** We considered the OWASP Top 10 for LLM Applications, particularly prompt injection and data leakage risks, when designing the assistant.

### 5.1.2 Hardware Standards

The system is designed to run on standard consumer-grade hardware (standard CPUs and entry-level GPUs) rather than requiring specialized computing infrastructure, since the RAG architecture avoids full model fine-tuning. Strict hardware requirements must be met by computing infrastructure that facilitates model training and inference.

**GPU and Computing Resources:** The minimum GPU needed to run a quantized open-weight LLM (Gemma, Qwen, or Llama, 7-9B parameters, 4 bit) is an NVIDIA GPU with 8 GB of VRAM. The recommended hardware is a GPU with 12-16 GB of VRAM, such as an NVIDIA [RTX 3060 12 GB / Tesla T4 16 GB]. During development, a free-tier cloud GPU (Google Colab/Kaggle) is used. During the user study, a university laboratory GPU or a local GPU is used to avoid session disconnects. Document embedding uses an open-source multilingual embedding model that runs on a standard CPU.

**Alternates Considered:**

- **High-end Cloud GPUs (AWS A100):** Scalable and fast, but recurring costs exceeded the FYDP budget, and the capacity is unnecessary for quantized inference.
- **CPU-only inference:** Removes the need for a GPU,but response times become too slow for a smooth user study.
- **Commercial LLM API:** Requires no local hardware, but adds recurring cost and gives less control over model versions, which affects reproducibility across treatments.

Because the RAG architecture avoids fine-tuning, entry-level GPUs are sufficient, and they are the most cost-effective option. Running quantized open-weight models on [lab / local] GPUs keeps the system low-cost and reproducible. All three treatments (A, B, C) use the same backend, so compute and latency stay identical and do not confound the NASA-TLX and SUS comparisons. Participants need only a web browser, so the system remains accessible on ordinary laptops and smartphones.

**Rationale for Selection:** As the RAG architecture avoids fine-tuning, entry-level GPUs are sufficient, and they are the most cost-effective option. Running quantized open-weight models on [lab / local] keeps the system low-cost and reproducible. All three treatments (A, B, C) use the same backend, so compute and latency stay identical and do not confound the NASA-TLX ans SUS comparisons. Participants need only a web browser, so the system remains accessible on ordinary laptops and smartphones rather than specialized devices.

**Memory and Storage Specifications:** A total 50 GB of local disk space is required for model weights (5-20 GB per model), the vector index, the Bangla QA dataset containing 299 QA pairs, and logs. The sysem requires at least 8 GB of system RAM. However, 16 GB is preferred so that the vector database, backend, and interface can run smoothly together.

**Computational Specifications:** It requires a multi-core GPU (4-8 cores), [Ubuntu 22.04 LTS/Windows 11], Python 3.10 or higher, and frameworks such as PyTorch, Hugging Face Transformers, [FAISS/Chroma], and [FastAPI/Flask] for the backend.

### 5.1.3 Communication Standards

**Data Interchange and Interoperability:** The three web-based interfaces (A, B, and C) communicate with the backend RAG system through a RESTful API over standard HTTP/HTTPS [built with FastAPI/Flask]. Secure data exchange between the frontend and the backend is formatted as JSON with UTF-8 encoding, which is required for the correct handling of Bengali legal text. HTTPS (TLS) is used during the user study to protect participant data in transit. Each response contains the generated answer, the cited source (Act, section, and page), and, for Treatment C, the retrieval confidence scores used for token-level highlighting and uncertainty warnings. All treatments use the same backend, protocol, and data format.

**Documentation and Reporting Standards:** Technical documentation follows IEEE Standard 1016-2009 for software design descriptions. The legal corpus is stored with structured metadata (Act, section, subsection, year, language). User study evaluation follows standardized NASA-TLX and System Usability Scale (SUS) instruments, and interaction logs (query, response, treatmentID, latency, and citation clicks) are stored as JSON under anonymized participant IDs.

**Version Control and Reproducibility:** Git repositories with detailed commit messages document all code changes, dataset revisions, and prompt versions. Experiments are run in Jupyter Notebooks or Google Colab with markdown explainations, and model versions and random seeds are fixed. Docker containerization is used to allow consistentand reproducible results across computing environments.

**Alternate Options:** Using WebSockets, gRPC streaming, or GraphQL subscriptions to maintain a persistent, bidirectional connection and stream responses token by token.

**Rationale for Selection:** REST over HTTP with JSON REST is simple, widely supported and works in any modern browser on ordinary laptops and smartphones without extra software. REST over HTTP with JSON also provides secure transmission of user queries, which may describe personal legal situations such as traffic incidents. Without additional client libraries, its simple implementation suits presenting all three interface treatments identically to a mixed-pool of laymen and legal expert participants. Returning each answer as one complete response lets Treatment C display citations, confidence bars, and token-level highlighting together with the text, and it avoids differences in reading pace caused by streaming. WebSockets and GraphQL add complexity that a fixed question-and-answer flow does not need, and gRPC is not natively supported in browsers. Using onebackend and one data formate for all three treatments keeps latency conistent across them, which prevents it from confounding the NASA-TLX and SUS comparisons.

## 5.2 Design Constraints

Developing an explainable legal assistant introduces several distinct design constraints that influenced architectural and operational decisions. Only the constraints directly relevant to RAG based legal assistants and its human-subjects user study are discussed below.

### 5.2.1 Economic Constraint

### 5.2.2 Environmental Constraint

### 5.2.3 Ethical Constraint

### 5.2.4 Health and Safety Constraint

### 5.2.5 Social Constraint

### 5.2.6 Political Constraints

### 5.2.7 Sustainability

For the project to be sustainable long-term, it must not depend on costly or hard-to-maintain infrastructure. By replying on open-source frameworks (PyTorch, Hugging Face Transformers, sentence-transformers, [LangChain / LlamaIndex], [FAISS / ChromaDB], and [FastAPI / Flask]) and open-weight models (Llama, Qwen, or Gemma), with Git version control and Docker containerization, future student teams can maintain, audit, and extend the system without licensing costs. The retrieval and interface pipeline is also designed to extend beyond the Road Transport Act, 2018 to other Bangladeshi statutes.

## 5.3 Cost Analysis

The development and evaluation of our explainable legal assistant require a structured financial plan to ensure feasibility. This section outlines the primary budget required for computation, expert validation of the dataset, and the user study; an alternate low-cost budget with its rationale, and the proposed revenue model for post-FYDP sustainability.

### 5.3.1 Proposed Budget

The primary budget focuses on acquiring stable computational resources for embedding, model benchmarking, and inference, and on paying legal-domain experts to validate the Bangla QA dataset and compensating the laymen and legal-expert participants of the 3×2 user study.

Table 5.1: Proposed Budget for the Explainable Legal Assistant

| **Category** | **Description** | **Estimated Cost (BDT)** |
|---|---|---|
| Computational Resources | Cloud GPU rental and subscriptions for embedding generation, model benchmarking, and stable inference during user study sessions | 30,000 |
| Data Annotation and Expert Validation | Compensation for legal-domain reviewers validating the Bangla QA dataset and ground-truth answers | 20,000 |
| User Study Incentives | Compensation for laymen and legal-expert participants across the 3×2 user study | 25,000 |
| Hosting and Infrastructure | Vector database hosting, web application hosting, and cloud storage | 15,000 |
| Miscellaneous | Contingency, printing, and internet data expenses | 5,000 |
| **Total** | | **95,000** |

### 5.3.2 Alternate Budget and Rationale

To account for strict financial constraints, an alternative budget is proposed that minimizes out-of-pocket expenses by leveraging existing and free resources.

Table 5.2: Alternate Low-Cost Budget

| **Category** | **Description** | **Estimated Cost (BDT)** |
|---|---|---|
| Computational Resources | Free-tier API credits and UIU lab GPUs/PCs for embedding generation and inference | 0 |
| Data Annotation and Expert Validation | Volunteer legal-studies faculty/students for dataset review | 0 |
| User Study Incentives | Volunteer participants recruited from the UIU community and personal networks | 0 |
| Hosting and Infrastructure | Free-tier web hosting and a locally hosted vector database | 0 |
| Miscellaneous | Contingency expenses | 4,500 |
| **Total** | | **4,500** |

**Rationale for Selection:** The proposed budget (Table 5.2) is highly preferred. Generating embeddings for the Road Transport Act, 2018 corpus, and running quantized open-weight LLMs for benchmarking and inference require significant VRAM, and free cloud tiers frequently interrupt long sessions and can disconnect during live user studies. Furthermore, paying legal-domain reviewers ensures that the Bangla QA dataset and its ground-truth labels are accurate, which is critical for reliably measuring hallucination detection and trust calibration. The alternate budget serves as a fallback but risks slower experimentation, a smaller validated dataset, and interrupted user study sessions.

### 5.3.3 Revenue Model

To ensure the long-term sustainability and scalability of the explainable legal assistant beyond the FYDP phase, the following revenue and value-creation model is proposed. As an academic research project, direct revenue creation is not the primary objective. However, the system has clear commercial and research potential:

- **Freemium Model for Citizens:** The core citation-backed question answering for the Road Transport Act, 2018, would be available for free to the general public, ensuring maximum social impact for citizens who cannot easily afford legal help due to cost, time, and lawyer accessibility. Premium features such as coverage of additional statues or saved query history could be gated behind a small subscription fee.

- **B2B Institutional Licensing:** The system could be licensed to legal-aid organizations, law firms, and government transport-related service bodies as a citizen-facing verified legal information tool,enabling users to independently cross-reference

AI-generated responses against the official statutory text.

- **API Provisioning:** The core dense retrieval pipeline and explainability module incorporating citation tracing and uncertainty scoring can be packaged as an API service and licensed to third-party legal-tech developers who need a verifiably grounded Bangla legal QA backend.
- **Research and Academic Value Creation:** Findings from the 3×2 calibration study could be published in peer-reviewed Human-Computer Interaction (HCI) and Explainable AI (XAI) venues. The expert-validated Bangla legal QA dataset and the design guidelines for trustworthy legal AI can serve as a foundation for future research and grants in Bangla legal NLP.

## 5.4 Complex Engineering Problem

### 5.4.1 Complex Problem Solving

In this section, provide a mapping with problem solving categories. For each mapping add subsections to put rationale (Use Table 5.3). For P1, you need to put another mapping with Knowledge profile and rational thereof.

Table 5.3: Mapping with complex problem solving.

| P1 Dept of Knowledge | P2 Range of Conflicting Requirements | P3 Depth of Analysis | P4 Familiarity of Issues | P5 Extent of Applicable Codes | P6 Extent of Stakeholder Involvement | P7 Interdependence |
|---|---|---|---|---|---|---|
| ✓ | ✓ | | | | | |

### 5.4.2 Engineering Activities

In this section, provide a mapping with engineering activities. For each mapping add subsections to put rationale (Use Table 5.4).

## 5.5 Summary

Table 5.4: Mapping with complex engineering activities.

| A1<br>Range of resources | A2<br>Level of Interaction | A3<br>Innovation | A4<br>Consequences for society and environment | A5<br>Familiarity |
|---|---|---|---|---|
| √ | √ | | | |

# Chapter 6

# Conclusion

[Must be present in FYDP-1 Report and also in Final Report. Might be incomplete in FYDP-1 Report.]

Every chapter should start with 1-2 sentences on the outline of the chapter.

## 6.1 Summary

## 6.2 Limitation

## 6.3 Future Work

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